



Louis Chataignier

Curriculum Vitae

Date of birth Dec 14, 1996
Nationality French
Email louis.chataignier@math.univ-toulouse.fr

Academic Background

- 2023–2026 **PhD student** at *Institut de Mathématiques de Toulouse*. Supervisors: Pascal Maillard and Michel Pain. Thesis title: *On particle trajectories and genealogy in branching Brownian motion and branching random walk*.
- 2022–2023 **Master in Advanced Mathematics** at *Université Lyon 1 – ENS de Lyon*, specializing in Probability and Statistics.
M2 thesis: *Additive martingales of the branching Brownian motion*. Supervisors: Pascal Maillard and Michel Pain. [arXiv:2407.20227](#).
- 2022 **Agrégation of Mathematics**, option Probability and Statistics. Admitted 24th in the national ranking.
- 2021–2022 **Master in General Mathematics** at *Université Lyon 1*.
M2 thesis: *Ordinary differential equations: examples of resolution and study of solutions in dimension 1 and 2*. Supervisor: Élise Fouassier.
- 2017–2018 **M1 in General Mathematics** at *Université Lyon 1*.
M1 thesis: *Formal study of the solvability of systems of partial differential equations*. Supervisor: Philippe Malbos.
- 2014–2017 **Bachelor's degree in General Mathematics** at *Université Lyon 1*.
2014 **Scientific Baccalauréat** at *Lycée du Forez, Feurs*.

Publications and preprints

- Jan 2026 *Upper moderate deviation probabilities for the maximum of a branching random walk*, Electron. Commun. Probab., with Lianghui Luo. ([arXiv:2601.08766](#)) *Preprint*.
- Nov 2025 *Upper moderate deviation probabilities for the maximum of branching Brownian motion*, Electron. Commun. Probab., 30, 1-14, <https://doi.org/10.1214/25-ECP737>. ([arXiv:2505.11363](#))
- Jul 2024 *Asymptotics of the overlap distribution of branching Brownian motion at high temperature*, with Michel Pain. ([arXiv:2407.21014](#)) *Preprint*.

Conferences and Seminars

- Jun 2026 Summer school **Particle systems and PDEs** at *Institut de Mathématiques de Toulouse*: attendance.
- May 2026 Conference **Discrete random structures II** at Będlewo (PL): attendance and presentation of a 20-minute talk.
- May 6, 2026 **PhD seminar** at *Institut Élie Cartan de Lorraine*, Nancy: presentation of a 1-hour talk (invited).
- Apr 2026 Annual conference of the **GDR Branchement** at Avignon: attendance.
- Dec 9, 2025 **Probability seminar** at *Beijing Institute of Technology* (CN): online presentation of a 1-hour talk (invited).
- Nov 25, 2025 **Probability seminar** at *Johannes Gutenberg-Universität Mainz* (DE): presentation of a 1-hour talk (invited).
- Nov 2025 **Franco-German doctoral college** on machine learning and probability at *Institut Agro Montpellier*: attendance and presentation of a 20-minute talk.
- Sep–Oct 2025 Colloquium **Jeunes probabilistes et statisticiens** at *CAES du CNRS*, Saint-Pierre-d’Oléron: attendance and presentation of a 25-minute talk.
- Apr 2025 Conference **Branching and persistence** at *Université d’Angers*: attendance and presentation of a 20-minute talk.
- Mar 2025 Workshop **Random discrete structures** at *Department of Earth Sciences of Oxford* (UK): attendance.
- Jan 2025 Annual conference of the **GDR Branchement** at Orsay: attendance.
- Jan 16, 2025 **Probability PhD students’ day** at *Institut de Mathématiques de Toulouse*: attendance and presentation of a 20-minute talk.
- Sep 2024 Conference **Non-local branching processes** at *Centre International de Rencontres Mathématiques*, Marseille: attendance and presentation of a 10-minute talk.
- Jul 2024 Probability summer school of **Saint-Flour** at *Ancien grand séminaire de Saint-Flour*: attendance and presentation of a 40-minute talk.
- Feb 2024 Spring school **Multiplicative chaos and cascades** at *Technical University of Darmstadt* (DE): attendance and presentation of a 10-minute talk.
- Jan 24, 2024 **Probability PhD students’ day** at *Institut de Mathématiques de Toulouse*: attendance and presentation of a 15-minute talk.
- Dec 21, 2023 **Christmas student seminar** at *Institut de Mathématiques de Toulouse*: attendance and presentation of a 15-minute talk.
- Nov 2023 1st conference of the **GDR Branchement** at *Institut de Mathématiques de Toulouse*: attendance.

Science Communication

- Jun 2024 **Classroom intervention to introduce probability theory** at *Collège Jacques Mauré*, Castelginest.
- Apr 2022 **Management of the Math@Lyon exhibition** at *Collège Marcel Pagnol*, Pierre-Bénite, with Régis Goiffon and Denis Serre.

Teaching

- 2025–2026 At *Université Toulouse III - Paul Sabatier*:
◦ Measure theory, L3, TD (56h).
- 2024–2025 At *Université Toulouse III - Paul Sabatier*:
◦ Introduction to probability theory, L2, TD/TP (38h).
◦ Measure theory, L3, TD (28h).
- 2023–2024 At *Université Toulouse III - Paul Sabatier*:
◦ Introduction to probability theory, L2, TD/TP (46h).
◦ Introduction to statistics, L2, lecture/TD (18h).
- 2022–2023 Weekly oral exams in preparatory classes at *Lycée du Parc*, Lyon.
- 2020–2021 Tutoring at *GoStudent*, Lyon.

Organizing Responsibilities

- Dec 2024– **Representative of PhD students and postdocs** on the laboratory council at *Institut de Mathématiques de Toulouse*.
- Spring 2025 **Co-organizer of a reading group** for the probability team of *Institut de Mathématiques de Toulouse*, with Jan-Luka Fatras.
- Autumn 2024 **Organizer of a reading group** for PhD students and postdocs in probability theory at *Institut de Mathématiques de Toulouse*.

Other Professional Experience

- 2019–2020 Kitchen porter at *SmokeWorks*, Cambridge (UK).
- 2018–2019 Kitchen porter at *Palégrié* ★, Corrençon-en-Vercors.

Skills

- Languages French (native), English (C2)
- Programming $\text{\LaTeX}$, Python
- Driving French driver's license